Lakshya JEE 2.0 2025

Maths

DPP: 12

Relations and Functions

Q1 If $f(x)$ is a polynomial function satisfying $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ and $f(4) = 65$, then $f(6)$ is:
(A) 36 (B) 37
(C) 216 (D) 217

Q2 If f is a function such that $f(0) = 2, f(1) = 3$, $f(x+2) = 2f(x) - f(x+1)$, then $f(5)$ is:
(A) -3 (B) -5
(C) 7 (D) 13

Q3 If $f\left(\frac{x+1}{x+2}\right) = \frac{x^2-1}{x^2+2}$, then find the value of $f(2)$.
(A) $\frac{1}{2}$
(B) $\frac{3}{2}$
(C) $\frac{8}{11}$
(D) $\frac{11}{8}$

Q4 If $f(x+10) + f(x+4) = 0$, there $f(x)$ is a periodic function with period.
(A) 2 (B) 4
(C) 6 (D) 12

Q5 If $2f\left(x^2\right) + 3f\left(\frac{1}{x^2}\right) = x^2 - 1$, then $f\left(x^2\right)$
(A) $\frac{1-x^4}{5x^2}$
(B) $\frac{1-x^2}{5x}$
(C) $\frac{5x^2}{1-x^4}$
(D) $\frac{3-x^2-2x^4}{5x^2}$

Q6 If $f(x) + 2f\left(\frac{1}{x}\right) = 3x, x \neq 0$, and $S = \{x \in R : f(x) = f(-x)\}$; then S
(A) Contains more than two elements
(B) Is an empty set
(C) Contains exactly one element
(D) Contains exactly two elements

Q7 Let a real valued function f satisfy $f(x+y) = f(x)\, f(y) \forall x, y \in R$ and $f(0) \neq 0$. Then:
$g\left(x\right) = \frac{f(x)}{1+[f(x)]^2}$ is:
(A) An even function
(B) An odd function
(C) Neither even nor odd function
(D) Periodic function

Q8 Let $g(x)$ be a polynomial function satisfying $g(x) \cdot g(y) = g(x) + g(y) + g(xy) - 2$ for all $x, y \in R$ and $g(1) \neq 1$. If $g(3) = 10$ then $g(5)$ equals:
(A) -24 (B) 16
(C) 26 (D) 34

Q9 If $f(x) = a\left(x^n + 3\right); f(1) = 12, f(3) = 36$, then $f(2)$ is equal to
(A) 21 (B) 18
(C) 24 (D) 27

Q10 Let $\sum_{k=1}^{10} f(a+k) = 16\left(2^{10} - 1\right)$, where the function f satisfies $f(x+y) = f(x)f(y)$ for all natural numbers x, y and $f(1) = 2$. Then the natural number a is:
(A) 2 (B) 3
(C) 16 (D) 4

Q11 If f is a polynomial function satisfying $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ and $f(2) = -3$,

then draw graph of $y = |f(-x)|$

Q12 If $f(x)$ is a polynomial function satisfying $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right), \forall x \in R$ and $- \{0\}$
$f(2) = -31$, then value of $f'(3)$ is:
(A) -324 (B) -405
(C) 405 (D) -108

Q13 If a, b be two fixed positive integers such that for all real x.

$$f(a+x) = b + \left(1 + b^3 - 3b^2(f(x)) + 3b(f(x))^2 - (f(x))^3\right)^{\frac{1}{3}},$$

then $f(x)$ is a periodic function with period
(A) a (B) $2a$
(C) b (D) $2b$

Q14 If $f(x) + 2f(1-x) = 6x (\forall x \in \mathrm{R})$, then the value of $\frac{3}{4}\left(\frac{f(8)}{f'(1)}\right)$ is equal to
(A) 5 (B) 5.5
(C) 6 (D) 6.5

Q15 Let $f : R \to R, g : R \to R$ be two functions given by $f(x) = 2x - 3, g(x) = x^3 + 5$. Then $(f \circ g)^{-1}(x)$ is equal to
(A) $(\frac{x+7}{2})^{1/3}$
(B) $(x - \frac{7}{2})^{1/3}$
(C) $(\frac{x-2}{7})^{1/3}$
(D) $(\frac{x-7}{2})^{1/3}$

Answer Key

Q1 (D)

Q2 (D)

Q3 (C)

Q4 (D)

Q5 (D)

Q6 (D)

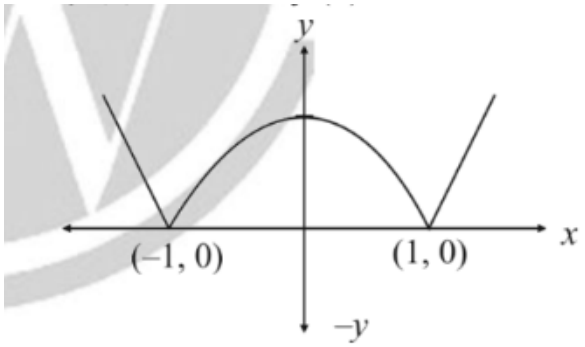
Q7 (A)

Q8 (C)

Q9 (A)

Q10 (B)

Q11



Q12 (B)

Q13 (B)

Q14 (B)

Q15 (D)

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